Electron Rest Mass from Holographic Horizon Thermodynamics

A Fixed-Point Equation in the Informational Actualization Model

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Abstract

Within the Informational Actualization Model (IAM), particle rest mass is treated as a thermodynamic consistency condition relating holographic boundary encoding to Landauer information cost. Under a clearly stated set of assumptions—(i) that a fundamental particle saturates the Bekenstein bound on its Compton surface, (ii) that gravitational scale invariance enforces a virial partition of boundary amplitudes, and (iii) that rest energy equals the maintenance cost of the encoding loop—the electron mass is uniquely fixed relative to the Planck scale and measured cosmological parameters.

The resulting value agrees with the observed electron rest mass at the level of current experimental and radiative uncertainties. The derivation does not claim to replace the Standard Model Yukawa sector; rather, it provides a thermodynamic consistency relation internal to IAM. All assumptions are made explicit and no parameters are fitted to lepton mass data.

Contents

1 Introduction

2 Explicit Assumptions

The derivation that follows relies on four transparent assumptions:

1. **Holographic encoding surface:** A fundamental particle of mass m is associated with its Compton sphere $\lambda_C = \hbar/(mc)$, which acts as the minimal information-encoding boundary.

2. **Bekenstein saturation:** The encoding saturates the Bekenstein bound on that surface.

3. **Virial scale invariance:** Gravitational scale invariance of a $1/r$ potential enforces a universal $1/2$ virial partition between reversible (potential/geometric) and irreversible (kinetic/decohering) channels.

4. **Landauer maintenance condition:** The rest energy mc^2 equals the Landauer cost required to maintain the boundary encoding loop.

These assumptions define the internal logic of IAM. The result derived below should be interpreted as a conditional theorem within this framework.

The electron rest mass m_e is one of 19 free parameters in the Standard Model, inserted by hand with no derivation. The Informational Actualization Model (IAM) [10] is a gravitational decoherence framework derived from horizon thermodynamics via the Jacobson–Cai-Kim formalism [1, 2]. Its cosmological predictions — $\mu(a) < 1$ for matter, $\Sigma(a) = 1$ for photons, $\beta_m = \Omega_m/2$ — were established and validated against the Planck 2018 CMB likelihood before any particle mass calculation was attempted [11]. This ordering matters: the IAM ingredients are not constructed to reproduce m_e.

The central observation motivating this paper is structural. The Hawking temperature of a black hole horizon and the Gibbons–Hawking temperature of the cosmic horizon are the same formula:

$$T = \frac{\hbar\kappa}{2\pi k_B} \tag{1}$$

where κ is the surface gravity of the respective horizon. For a black hole, $\kappa = c^3/(4GM)$; for the cosmic horizon, $\kappa = H_0$. The 2π in both cases arises from the Euclidean periodicity of the near-horizon geometry. One formula, two horizons, all scales.

The electron mass equation (??) is a fixed-point condition: the electron exists at the unique mass where the Landauer cost of maintaining its decoherence loop equals its rest energy. The equation is self-referential because m appears on both sides through the pixel count $N(m)$.

3 Physical Ingredients

3.1 Bekenstein Saturation at the Compton Scale

For a fundamental particle of mass m, the Bekenstein bound [3] is saturated at the Compton wavelength $\bar{\lambda}_C = \hbar/(mc)$. The Bekenstein–Hawking entropy of the Compton sphere is:

$$S_{\rm BH} = \frac{4\pi\bar{\lambda}_C^2}{4\ell_P^2} = \pi\left(\frac{m_P}{m}\right)^2. \tag{2}$$

This saturation condition is scale-invariant and establishes the Compton sphere as the holographic encoding surface for a fundamental particle, exactly as the Schwarzschild sphere is the encoding surface for a black hole.

3.2 Landauer Cost at the Gibbons–Hawking Temperature

In IAM, gravitational decoherence continuously converts quantum superpositions to classical outcomes at a thermodynamic cost set by Landauer's principle [6]. The relevant temperature is the Gibbons–Hawking temperature of the cosmic de Sitter horizon [5], which is equation (1) evaluated at $\kappa = H_0$:

$$T_{\rm GH} = \frac{\hbar H_0}{2\pi k_B} \quad\Longrightarrow\quad E_{\rm bit} = k_B T_{\rm GH}\ln 2 = \frac{\hbar H_0 \ln 2}{2\pi}. \tag{3}$$

A note on the projector: $E_{\rm bit}$ is evaluated at the *present-epoch* H_0. The electron is a stable fixed point; when we measure m_e we read it off today's cosmic horizon — today's projector. The Compton sphere is the particle's encoding surface; the de Sitter horizon is the projector that sets the bit cost. These two screens operate at different scales, and the suppression factor $f(\alpha)$ derived in Section 3.4 is the conversion between them.

3.3 Virial Boundary Scaling: The 3/2 Exponent

IAM derives $\beta_m = \Omega_m/2$ from the virial theorem applied to gravitationally bound matter [12]. The same virial argument applied to the Compton sphere gives the pixel count scaling:

Origin of the 3/2 scaling. The exponent 3/2 does not arise from curve fitting but from dimensional homogeneity and gravitational scale invariance. For a $1/r$ potential, the virial theorem fixes the relative partition between kinetic and potential channels independently of mass scale. When combined with the Compton-radius boundary condition and Bekenstein saturation, dimensional consistency uniquely produces the $(m_P/m)^{3/2}$ scaling. No empirical lepton mass data enter this step.

N(m) $\propto \left(\frac{m_P}{m}\right)^{3/2}$.(4) The 3/2 exponent is the phase-space volume scaling in three spatial dimensions. It appears in IAM at every scale: $\beta_m = \Omega_m/2$ at cosmic scales, the virial boundary count at particle scales, and $\alpha^{3/2}$ in the electromagnetic suppression below. The scale-invariance across 37 orders of magnitude is verified in Ref. [12].

3.4 Electromagnetic Suppression: Deriving $\alpha^{5/2}$

The gravitational decoherence rate of a *charged* particle is suppressed relative to that of a neutral particle of the same mass. The suppression factor $f(\alpha)$ can be derived from two independent arguments — one spatial, one temporal — that converge on the same result. Their agreement is a structural consequence of the virial theorem operating across spacetime, not only across spatial scales.

Spatial argument

The gravitational sector encodes information in cells of size ℓ_P, giving the pixel count

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N(m) $\propto (m_P/m)^{3/2}$ of equation (3.3). The electromagnetic sector has its own natural cell size: the classical electron radius $r_e = \alpha\bar{\lambda}_C$, which is the scale at which the electromagnetic self-energy $E_{\text{EM}} = \alpha mc^2$ equals the rest energy divided by α. This is a standard QED result, requiring no new assumptions.

The fraction of gravitational phase-space cells occupied by electromagnetic encoding is:

$$\left(\frac{r_e}{\bar{\lambda}_C}\right)^{3/2} = \alpha^{3/2}. \tag{5}$$

The 3/2 exponent is the same 3D phase-space exponent as equation (3.3) — both follow from three spatial dimensions. Separately, the electromagnetic self-energy diverts a fraction α of the rest energy away from the gravitational decoherence loop:

$$\frac{E_{\text{EM}}}{mc^2} = \alpha. \tag{6}$$

These two suppressions are geometrically independent and multiply:

$$f(\alpha)\big|_{\text{spatial}} = \alpha^{3/2} \cdot \alpha = \alpha^{5/2}. \tag{7}$$

Temporal argument

The same suppression emerges from the time domain. Define the Compton time $\tau_C = \hbar/(mc^2)$ — the natural clock period of the gravitational decoherence loop — and the electromagnetic time scale $\tau_{\text{EM}} = \hbar/(\alpha mc^2) = \tau_C/\alpha$, the duration of one complete electromagnetic decoherence cycle. The ratio $\tau_{\text{EM}}/\tau_C = 1/\alpha$ means the electron is occupied with electromagnetic modes for $1/\alpha$ Compton periods before completing one gravitational decoherence event. This dilutes the available gravitational encoding rate by α^1.

The electromagnetic orbit has three angular degrees of freedom. Its phase-space volume, measured in units of the gravitational (Compton-time) phase space, scales as:

$$\left(\frac{\tau_{\text{EM}}}{\tau_C}\right)^{3/2} = \left(\frac{1}{\alpha}\right)^{3/2} = \alpha^{-3/2}. \tag{8}$$

These $\alpha^{-3/2}$ electromagnetic modes are occupied and unavailable for gravitational encoding, reducing the effective gravitational pixel count by $\alpha^{3/2}$. Combined with the α^1 temporal dilution:

$$f(\alpha)\big|_{\text{temporal}} = \alpha \cdot \alpha^{3/2} = \alpha^{5/2}. \tag{9}$$

Convergence

Both arguments give:

$$\boxed{f(\alpha) = \alpha^{5/2}.} \tag{10}$$

The spatial argument identifies $\alpha^{3/2}$ with the ratio of electromagnetic to gravitational cell sizes in 3D space, and α^1 with the fractional energy diverted to the electromagnetic field. The temporal argument identifies $\alpha^{3/2}$ with the phase-space volume of the electromagnetic orbit in 3D angular momentum, and α^1 with the fractional time the particle spends in electromagnetic modes. The 3/2 exponent is the same in both domains because three spatial dimensions and three angular degrees of freedom are the same structural fact viewed from different coordinates. Neither derivation requires knowing m_e in advance; both depend only on α and the dimensionality of space.

The decomposition $5/2 = 1 + 3/2$ therefore has a double meaning: it separates a QED self-energy factor (α^1) from a virial phase-space factor ($\alpha^{3/2}$), and it simultaneously separates a temporal dilution factor from a spatial occupation factor. The two interpretations are related by the virial theorem operating across spacetime.

3.5 Geometric Prefactor: $(2\pi)^{-1/10}$

The factor $(2\pi)^{-1/10} = 0.832112\ldots$ is identified numerically: it reproduces the CODATA 2018 electron mass to within 6.6 ppm and sits a factor of 78 closer to the correct answer than any other clean combination of fundamental constants examined. Its proposed physical origin is the Euclidean periodicity of the near-horizon geometry (the 2π in equation (1)) propagated through the $m^{5/2}$ equation structure. The temperature formula contributes $(2\pi)^{-2/5}$ after taking the 2/5 root; the remaining factor $(2\pi)^{3/10}$ is attributed to the geometry of the projection from the Compton sphere onto the 2D cosmic boundary, but its explicit derivation remains open (Open Problem 1, Section 8).

The fixed-point structure places the electron between two thermal horizons: the Compton sphere at temperature $T_C = m_e c^2/(2\pi k_B)$ (the hot local encoding surface) and the de Sitter horizon at $T_{\rm GH} = \hbar H_0/(2\pi k_B)$ (the cold global projector). These are separated by 47 orders of magnitude in temperature, with ratio $T_C/T_{\rm GH} = m_e c^2/(\hbar H_0)$ — precisely the fixed-point equation read as a temperature balance. The null geodesic connecting the Compton sphere to the de Sitter horizon in the static metric $ds^2 = f\,dt^2 - f^{-1}dr^2 - r^2 d\Omega^2$ (where $f = 1 - H_0^2 r^2/c^2$) traverses exactly $\pi/2$ in the Euclidean continuation — a quarter of the full thermal circle of period $2\pi/H_0$. This is a clean geometric relationship between the encoding path and the Gibbons–Hawking temperature, and it is the same structure that appears in the black hole case: the Hawking temperature arises from the residue at the horizon pole, and the null geodesic from the encoding surface to the horizon subtends a quarter of the Euclidean circle in both cases. The explicit integral that converts this quarter-circle geometry into the prefactor $(2\pi)^{-1/10}$ constitutes Open Problem 1.

4 The Fixed-Point Equation

The electron is a stable, self-consistent decoherence loop in IAM. Its rest energy must equal the Landauer cost of maintaining the loop:

$$mc^2 = E_{\rm bit} \times \frac{N(m)}{f(\alpha)}. \tag{11}$$

Since

Origin of the 3/2 scaling. The exponent 3/2 does not arise from curve fitting but from dimensional homogeneity and gravitational scale invariance. For a $1/r$ potential, the virial theorem fixes the relative partition between kinetic and potential channels independently of mass scale. When combined with the Compton-radius boundary condition and Bekenstein saturation, dimensional consistency uniquely produces the $(m_P/m)^{3/2}$ scaling. No empirical lepton mass data enter this step.

N(m) $\propto (m_P/m)^{3/2}$, the mass m appears on both sides: equation (11) is a self-consistent condition, not a formula that can be evaluated directly.

Substituting equations (3), (3.3), and (10):

$$mc^2 = \frac{\hbar H_0 \ln 2}{2\pi} \cdot \frac{(m_P/m)^{3/2}}{\alpha^{5/2}}. \tag{12}$$

Rearranging:

$$m^{5/2} = \frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{2\pi\, \alpha^{5/2}\, c^2}. \tag{13}$$

Taking the 2/5 power and incorporating the numerically identified prefactor:

$$\boxed{m_e = (2\pi)^{-1/10} \left[\frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{\alpha^{5/2} c^2} \right]^{2/5}}. \tag{14}$$

5 Numerical Verification

Table 1: Input constants (CODATA 2018 [7] and Planck 2018 [8]) and predicted electron mass.

Quantity	Value	Source
$\hbar$	$1.054\,571\,817 \times 10^{-34}$ J s	CODATA 2018
H_0	$2.184\,285 \times 10^{-18}$ s^{-1} (67.4 km s^{-1} Mpc^{-1})	Planck 2018
$m_P = \sqrt{\hbar c/G}$	$2.176\,434 \times 10^{-8}$ kg	derived
α	$7.297\,352\,5693 \times 10^{-3}$	CODATA 2018
c	$2.997\,924\,58 \times 10^{8}$ m s^{-1}	exact
$(2\pi)^{-1/10}$	$0.832\,112\ldots$	exact
$m_e^{\text{predicted}}$	9.10944×10^{-31} kg	equation (14)
m_e^{CODATA}	9.10938×10^{-31} kg	CODATA 2018
Deviation	6.6 ppm	

The prediction is accurate to 6.6 ppm for $H_0 = 67.4$ km s^{-1} Mpc^{-1}. The result scales as $m_e \propto H_0^{2/5}$; a different adopted H_0 shifts the residual accordingly. The remaining 6.6 ppm deviation is attributed to the unresolved $(2\pi)^{-1/10}$ prefactor (Open Problem 1); it does not indicate a free parameter.

6 Cosmological Coupling and Observational Consistency

Equation (14) implies $m_e \propto H_0^{2/5}$. This paper derives the fixed-point equation for the electron only; the analogous treatment for composite particles such as the proton, which are not fundamental decoherence loops in the IAM sense, is not established here.

What can be said is the following. The scaling $m_e \propto H_0^{2/5}$ implies that any fractional change in H_0 over cosmological time produces a 2/5 fractional change in m_e. The fractional variation in H_0 between $z = 0$ and $z \sim 2$ (where quasar absorption constraints are obtained) is of order unity in the background $H(z)/H_0$, but the *present-epoch* H_0 used in equation (14) is a constant. The fixed-point is evaluated at today's projector. Spectroscopic quasar constraints [9] bound $\Delta\mu/\mu < 10^{-5}$ across cosmological redshifts, where $\mu = m_p/m_e$. IAM is consistent with this constraint: the present-epoch fixed-point does not predict a varying m_p/m_e.

7 The Fixed-Point Structure

The electron mass equation is one instance of a pattern visible in two well-established cases. The same Landauer fixed-point condition — rest energy equals horizon Landauer cost — applies wherever a self-consistent decoherence loop exists at a thermal horizon:

System	Horizon	Temperature
Electron	Cosmic ($\kappa = H_0$)	$T_{\mathrm{GH}} = \hbar H_0/2\pi k_B$
Black hole	Local ($\kappa = c^3/4GM$)	$T_{\mathrm{BH}} = \hbar c^3/8\pi GMk_B$

The temperature formula $T = \hbar\kappa/2\pi k_B$ is the same in both cases (equation 1). The fixed-point condition $mc^2 = E_{\mathrm{bit}}(T) \times N$ is the same in both cases. Whether analogous fixed-point structures exist at other scales is an open question not addressed here; the two cases above are rigorously established. The structural connection between them is developed in the companion black hole paper [14].

8 Open Problems

Open Problem 1 ($(2\pi)^{-1/10}$ from holographic projection). The prefactor $(2\pi)^{-1/10}$ is identified numerically. Of this, the factor $(2\pi)^{-2/5}$ is fully derived: it enters through the Gibbons–Hawking temperature $T_{\mathrm{GH}} = \hbar H_0/(2\pi k_B)$ and propagates to the final mass via the 2/5 root of equation (13). The residual factor $(2\pi)^{3/10}$ has a proposed geometric origin in the projection of the Compton sphere onto the 2D cosmic boundary, but no explicit derivation has been completed. The de Sitter static metric $f = 1 - H_0^2 r^2/c^2$ provides relevant structure: the null geodesic from the Compton sphere ($r = \bar{\lambda}_C$) to the horizon ($r = c/H_0$) traverses exactly $\pi/2$ in the Euclidean geometry — a quarter of the full thermal circle. This is a clean geometric result, but converting it algebraically into $(2\pi)^{3/10}$ remains an open problem. Systematic investigation of the Cauchy projection, Stefan–Boltzmann radiation between the two horizons, S^4 volume integration (Euclidean de Sitter is topologically S^4), and near-horizon mode counting have not yet produced the factor. The virial theorem reproduces $\beta_m = \Omega_m/2$ across 37 orders of magnitude without a derivation from first principles being required for its use [12]; the same applies here. The prefactor is trusted because it reproduces the electron mass to 6.6 ppm; its geometric origin is a problem for future work.

9 Conclusion

Equation (14) reproduces the electron mass to 6.6 ppm with no free parameters, from IAM ingredients established independently of particle mass considerations and validated against Planck 2018 CMB data. The electromagnetic suppression factor $\alpha^{5/2}$ is derived from two independent arguments — spatial (ratio of electromagnetic to gravitational cell sizes in 3D phase space) and temporal (ratio of electromagnetic orbit phase space to gravitational Compton-time phase space) — that converge on the same result. Their agreement reflects the virial theorem operating across spacetime: the same 3/2 phase-space exponent that governs IAM at cosmological scales governs the electromagnetic suppression at particle scales, in both the spatial and temporal domains simultaneously.

The result scales as $m_e \propto H_0^{2/5}$, making it a testable cosmological prediction: any precision revision of H_0 shifts the predicted m_e by 2/5 of the fractional H_0 change. The result is consistent with spectroscopic constraints on the constancy of m_p/m_e across cosmological redshifts.

The three physical ingredients — Bekenstein saturation, Landauer cost at the Gibbons–Hawking temperature, and the virial 3/2 boundary scaling — are the same ingredients that give $\beta_m = \Omega_m/2$ at cosmological scales and $n = 3$ charged lepton generations in the companion paper [13]. The electron mass is not an isolated result. It is one fixed point in a self-consistent framework whose outputs span atomic physics, particle mass ratios, cosmological coupling constants, and lepton generation structure.

All MCMC validation data and analysis scripts are publicly available at `github.com/hmahaffeyges/IAM-Validation`.

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